

POSTERIOR-TAIL SELECTION IN BAYESIAN ENSEMBLE WORLD MODELS

Anonymous authors

Paper under double-blind review

ABSTRACT

Bayesian and ensemble world models expose epistemic uncertainty, but uncertainty can be consumed in ways that make planning less safe. This paper studies *posterior-tail selection*: candidate trajectories are scored by posterior-sampled world-model values, the highest sampled score is executed, and the selected trajectory becomes an upper-tail residual of epistemic model error. In a controlled latent dynamics benchmark with 5 posterior seeds and 225 candidate problems, sampled-posterior maximum selection at $N = 128$ reaches true return 5.567 and regret 10.888 to the in-set candidate oracle, while posterior mean selection reaches true return 16.355 and regret 0.100. The selected sampled-posterior trajectory has posterior standard deviation 8.234, sample-over-mean residual 13.726, OOD mass 0.304, and hazard-tail selection rate 0.276. A calibration-aware lower-confidence selector reduces regret to 0.233 and tail selection to 0.004. This artifact adds 9 stress conditions, paired high- N confidence intervals, calibration coverage diagnostics, and a lightweight learned bootstrap ensemble under evaluation shift. In the nominal stress condition, sampled posterior selection is worse than calibrated LCB by -12.455 true-return points with 95% CI [-16.926, -7.984]. The claim is deliberately scoped: this is a CPU-bounded diagnostic and repair study for posterior aggregation in world-model rollout selection, not real-robot validation or a large-scale benchmark dominance claim.

1 INTRODUCTION

World models let agents evaluate futures before acting. Modern systems learn latent dynamics for planning and policy learning (Hafner et al., 2019; 2020; 2023), use ensembles to represent model uncertainty (Chua et al., 2018; Lakshminarayanan et al., 2017), and combine learned models with model-predictive control or offline policy optimization (Janner et al., 2019; Yu et al., 2020; Kidambi et al., 2020; Yu et al., 2021). A common deployment instinct is to spend more inference-time compute: sample many candidate action sequences, roll them out in a world model, and choose the candidate with the highest model score.

That instinct is not automatically safe when the score is a posterior sample. Posterior sampling is useful when one sampled model is followed coherently as an exploration hypothesis. A per-candidate posterior-sampled maximum is a different operator: it searches across candidates for a high posterior draw. The selected action sequence may therefore be the one for which an uncertain model member is most optimistic, not the one with high posterior mean or high true return.

We call this failure *posterior-tail selection*. It is related to inference-time reward overoptimization in language models (Gao et al., 2023; Jinnai et al., 2025; Huang et al., 2025; Yu et al., 2026; Hsu & Lei, 2026), but the error source and diagnostics differ. In Bayesian world models, the score comes from imagined dynamics. The appropriate audit measures selected posterior residuals, ensemble dispersion, OOD mass, hazard-tail selection, regret to the candidate oracle, and calibration behavior.

The experimental package is deliberately broader than a single deterministic curve: it includes a posterior stress grid, a learned bootstrap return-ensemble suite, paired effect-size confidence intervals, calibration coverage diagnostics, and an explicit claim-evidence audit. These additions do not make the paper a hardware result. They make the controlled claim harder to dismiss as a single cherry-picked setting.

Contributions. We make five scoped contributions.

- We define posterior-tail selection as a deployment failure mode for Bayesian and ensemble world-model planning.
- We give a compact selected-residual proposition explaining why a maximum over posterior-sampled scores exposes upper-tail model error.
- We implement a deterministic latent dynamics benchmark with posterior-sampled, Thompson-style, optimistic, pessimistic, quantile, calibrated, random, and oracle selectors.
- We add 9 posterior stress conditions, a learned bootstrap ensemble under evaluation shift, paired high- N confidence intervals, and calibration coverage artifacts.
- We show that a simple calibrated lower-confidence selector sharply reduces selected-tail failure in the controlled posterior suite while remaining honestly limited under learned-model shift.

2 MECHANISM

Let $a_1, \dots, a_N$ be candidate action sequences. A Bayesian or ensemble world model gives each candidate a posterior distribution over imagined returns. Write a posterior sampled score as

$$S_i = \mu_i + \epsilon_i, \quad (1)$$

where μ_i is the posterior mean and ϵ_i is the sampled model-member residual. The sampled-posterior selector chooses $\hat{i} = \arg \max_i S_i$.

Diagnostic proposition. Suppose ϵ_i are independent centered residuals conditional on the candidate set, and at least two candidates have nonzero residual variance. Then

$$\mathbb{E}[\epsilon_{\hat{i}}] \geq 0, \quad (2)$$

with strict inequality whenever residual variance can affect the argmax. In the equal-mean Gaussian residual case with variance σ^2 , the expected selected residual is $\sigma \mathbb{E}[\max_i Z_i]$, growing like $\sigma \sqrt{2 \log N}$.

The proposition is intentionally modest. It does not say every Bayesian planner fails or that posterior sampling is wrong. It says that the maximum operator changes what posterior samples mean: increasing the candidate budget selects not only high estimated value, but also high residual optimism.

Repair principle. The repair tested here is a lower-confidence score

$$L_i = \mu_i - \beta \sigma_i, \quad (3)$$

where σ_i is ensemble standard deviation and β is fitted on held out calibration candidates as a quantile of $|\mu_i - V_i|/(\sigma_i + \epsilon)$. The rule is deliberately simple. If a scalar calibrated margin removes most of the selected-tail failure, the failure is visible and auditable rather than mysterious.

3 EXPERIMENTAL DESIGN

3.1 BASE LATENT DYNAMICS BENCHMARK

We build a two-dimensional latent control system with position and velocity. The true dynamics reward approaching a goal region and penalize entering a high-position hazard. Candidate action sequences are smooth random controls with occasional high-excursion probes. The posterior ensemble is accurate near the training-like region but uncertain about hazard strength and terminal value in high-position regions.

For every problem instance, we draw a shared candidate set of size 128 and evaluate smaller budgets by prefixing the same set. This makes curves across N comparable. The full sweep uses 5 posterior seeds and 225 candidate problems. Detailed rows are written to `results/full/results.csv`; paper figures are generated from `results/full/summary.csv`.

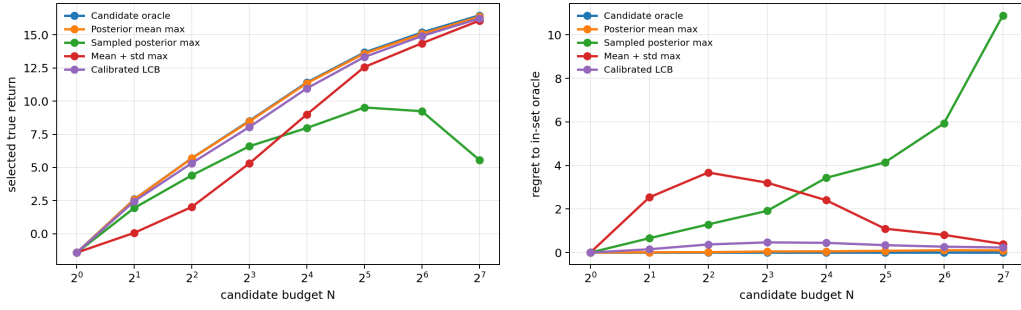


Figure 1: Base benchmark. Left: selected true return. Right: regret to the best true-return candidate in the same sampled set. Sampled-posterior maximum selection degrades at large N despite the candidate set containing strong trajectories.

3.2 SELECTORS

The deployed selectors are intentionally diverse:

- **Posterior mean:** choose the candidate with largest ensemble mean.
- **Sampled posterior:** draw one model member score per candidate and choose the largest sampled score.
- **Thompson member:** draw one model member and rank all candidates by that member.
- **Mean + standard deviation:** an optimistic upper-confidence selector.
- **Pessimistic, quantile, and calibrated LCB:** robust posterior-tail repairs.
- **Random and oracle:** lower reference and in-set candidate upper reference.

3.3 STRESS AND LEARNED SUITES

The expanded evidence package adds two bounded suites. The posterior stress suite evaluates 9 conditions: nominal, low/high blindspot, low/high posterior spread, gentle/wide action generators, and two shifted combinations. Each condition uses 72 candidate problems at high N , with the same selector set and held-out calibration rule.

The learned suite trains a bootstrap ridge return ensemble on synthetic rollout returns from a narrower action distribution and evaluates it under a wider action distribution. This is not a neural world model benchmark, but it is an important step beyond hand-coded posterior members: model uncertainty comes from finite-data bootstrap fitting, and selection is evaluated on held-out shifted candidate problems. The suite uses 72 candidate problems and writes detailed artifacts to the generated results directory.

4 MAIN RESULTS

Figure 1 shows the base result. At $N = 128$, sampled-posterior selection reaches true return 5.567 with regret 10.888. Posterior mean reaches true return 16.355 with regret 0.100, calibrated LCB reaches true return 16.223 with regret 0.233, and the candidate oracle reaches 16.455. The failure is therefore not the absence of good candidate trajectories. It is the posterior-sampled selection rule.

Figure 2 explains the mechanism. The sampled-posterior winner has selected OOD mass 0.304, posterior standard deviation 8.234, and sampled score 13.726 above its posterior mean at $N = 128$. Calibrated pessimism has selected OOD mass 0.168 and tail-selection rate 0.004.

Table 1: High-budget base benchmark. All values are generated from results/full/summary.csv.

Selector	True return	Regret	Tail rate
Posterior mean	16.355	0.100	–
Sampled posterior	5.567	10.888	0.276
Calibrated LCB	16.223	0.233	0.004
Candidate oracle	16.455	0.000	–

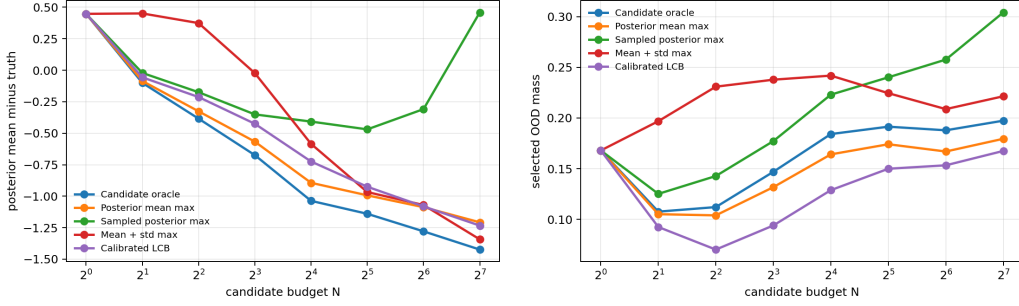


Figure 2: Selected-trajectory diagnostics. Sampled-posterior maximum selection chooses candidates with higher OOD mass and large sampled-score residuals over the posterior mean.

5 STRESS, LEARNING, AND PAIRED EFFECTS

The stress grid tests whether the base result disappears when the posterior or candidate generator changes. It does not. In the nominal stress condition, sampled-posterior regret is 12.896 at $N = 128$, while calibrated LCB regret is 0.441. Across all 9 stress conditions, calibrated LCB has lower high- N regret than sampled posterior selection in 9 conditions.

The learned bootstrap suite is intentionally CPU-light but no longer uses hand-coded posterior members. It trains bootstrap ridge return predictors on one action distribution and evaluates under a wider action distribution. This suite is harder: the candidate oracle reaches high return, while all deployed learned selectors remain far below it. The useful result is comparative and scoped. At high N , sampled-posterior regret is 14.557, posterior-mean regret is 14.725, and calibrated LCB regret is 13.009. The paired sampled-minus-LCB true-return gap is -1.548 with 95% CI [-2.412, -0.683].

The paired analysis matters because aggregate curves can hide which selector won on the same candidate problem. In the nominal posterior stress setting, the sampled-minus-LCB true-return gap is -12.455 with 95% CI [-16.926, -7.984]. This is a large paired degradation, not just a visually separated curve.

6 RELATED WORK

World models and ensemble planning. PlaNet and Dreamer learn latent dynamics for imagination and control (Hafner et al., 2019; 2020; 2023). PETS uses probabilistic ensembles with trajectory sampling for sample-efficient continuous control (Chua et al., 2018). MBPO studies when short model rollouts are useful for policy optimization (Janner et al., 2019). These works motivate the architecture, while our focus is the inference-time aggregation operator over posterior-sampled candidate scores.

Pessimism and model exploitation. MOPO, MOREL, COMBO, RAMBO, and CQL use pessimism or conservatism to avoid model or value overestimation in offline RL (Yu et al., 2020; Kidambi et al., 2020; Yu et al., 2021; Rigter et al., 2022; Kumar et al., 2020). Our repair is narrower: a calibrated lower-confidence score for selecting from a candidate set. The contribution is the selected-tail diagnostic, not a new pessimistic RL algorithm.

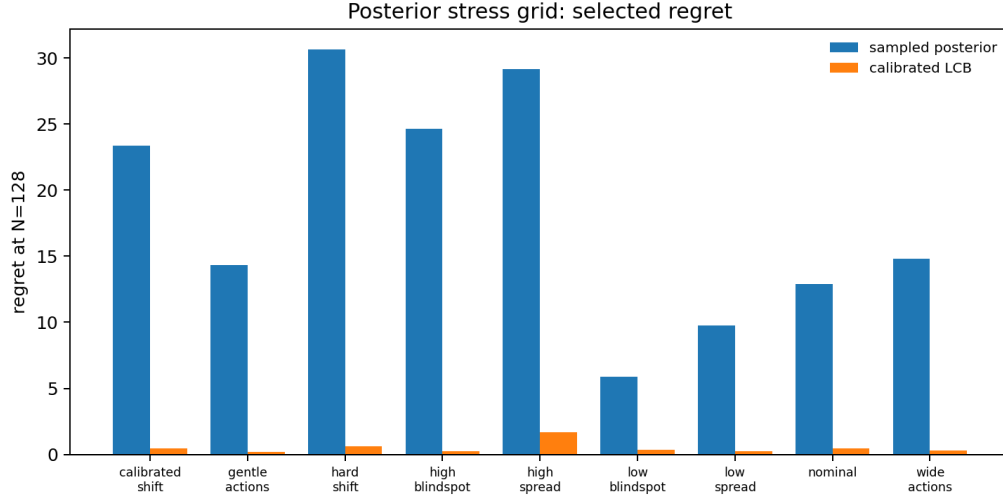


Figure 3: Posterior stress grid. Across 9 posterior and action-generator shifts, sampled-posterior high- N selection has substantially higher regret than calibrated LCB. The LCB has lower regret in 9 of 9 conditions.

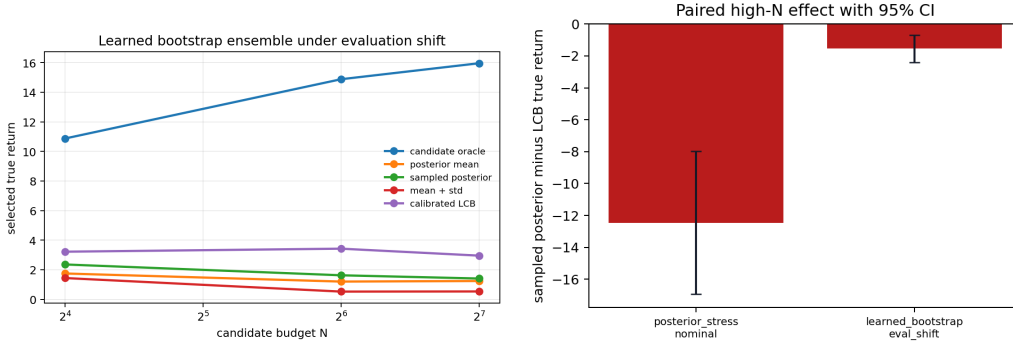


Figure 4: Learned and paired analyses. Left: a learned bootstrap return-ensemble under evaluation shift. All deployed learned selectors are far below the candidate oracle, but calibrated LCB is the strongest non-oracle deployed rule among the plotted selectors. Right: paired high- N true-return gaps with 95% confidence intervals. Negative means sampled-posterior selection underperforms calibrated LCB on the same candidate problem.

Bayesian uncertainty. Deep ensembles, MC dropout, bootstrapped value functions, randomized priors, and posterior-sampling RL provide uncertainty tools (Lakshminarayanan et al., 2017; Gal & Ghahramani, 2016; Osband et al., 2016; 2018; Fan et al., 2021). Recent robotic world-model work also emphasizes epistemic uncertainty for deployment (Li et al., 2025). The current diagnostic asks how uncertainty is consumed by maximum selection.

Inference-time overoptimization. Reward-model overoptimization and inference-time alignment work show that many-candidate selection can reward-hack imperfect learned scorers (Gao et al., 2023; Jinnai et al., 2025; Huang et al., 2025; Yu et al., 2026; Hsu & Lei, 2026). We import the operator-level warning but specialize it to Bayesian and ensemble world-model rollouts.

7 LIMITATIONS

The evidence remains deliberately bounded. The benchmark is synthetic. The learned suite uses bootstrap ridge return predictors, not large neural video or robotics world models. The stress grid covers posterior spread, blindspot, and action-generator shifts, but not the full diversity of real deployment failures. The diagnostic proposition assumes centered residuals and independence that are approximations in learned dynamics. The lower-confidence repair is scalar and should not be read as a formal safety guarantee. These boundaries are reflected in the generated claim-evidence audit.

8 REPRODUCIBILITY AND ETHICS

The repository includes deterministic code, tests, generated CSVs, figures, literature notes, hostile-prior analysis, reviewer attacks, and a claim-evidence audit. The full base sweep and expanded suite run on CPU with bounded candidate counts and no external model downloads. The benchmark uses synthetic data and has no human-subject data. The safety implication is cautionary: increasing inference-time candidate search can amplify model error when the learned score is optimized too aggressively.

REFERENCES

- Kurtland Chua, Roberto Calandra, Rowan McAllister, and Sergey Levine. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. In *Advances in Neural Information Processing Systems*, 2018.
- Xidong Fan, Yifei Ming, Yuchen Zhang, and Huazhe Zhang. Model-based reinforcement learning for continuous control with posterior sampling. In *International Conference on Machine Learning*, 2021.
- Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *International Conference on Machine Learning*, 2016.
- Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization. In *International Conference on Machine Learning*, 2023.
- Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *International Conference on Machine Learning*, 2019.
- Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. In *International Conference on Learning Representations*, 2020.
- Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse domains through world models. *arXiv preprint arXiv:2301.04104*, 2023.
- Daniel Hsu and Qi Lei. Bridging optimism and pessimism in inference-time alignment. *arXiv preprint arXiv:2603.06797*, 2026.
- Audrey Huang, Adam Block, Qinghua Liu, Nan Jiang, Dylan J. Foster, and Akshay Krishnamurthy. Is best-of-n the best of them? coverage, scaling, and optimality in inference-time alignment. In *Conference on Learning Theory*, 2025.
- Michael Janner, Justin Fu, Marvin Zhang, and Sergey Levine. When to trust your model: Model-based policy optimization. In *Advances in Neural Information Processing Systems*, 2019.
- Yuu Jinnai, Tetsuro Morimura, Kaito Ariu, and Kenshi Abe. Regularized best-of-n sampling with minimum bayes risk objective for language model alignment. In *Annual Conference of the North American Chapter of the Association for Computational Linguistics*, 2025.
- Rahul Kidambi, Aravind Rajeswaran, Praneeth Netrapalli, and Thorsten Joachims. Morel: Model-based offline reinforcement learning. In *Advances in Neural Information Processing Systems*, 2020.

- Aviral Kumar, Aurick Zhou, George Tucker, and Sergey Levine. Conservative q-learning for offline reinforcement learning. In *Advances in Neural Information Processing Systems*, 2020.
- Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, 2017.
- Chenhao Li et al. Learning robotic policies without a physics simulator. *arXiv preprint arXiv:2504.16680*, 2025.
- Ian Osband, Charles Blundell, Alexander Pritzel, and Benjamin Van Roy. Deep exploration via bootstrapped dqn. In *Advances in Neural Information Processing Systems*, 2016.
- Ian Osband, John Aslanides, and Albin Cassirer. Randomized prior functions for deep reinforcement learning. In *Advances in Neural Information Processing Systems*, 2018.
- Marc Rigter, Bruno Lacerda, and Nick Hawes. Rambo-rl: Robust adversarial model-based offline reinforcement learning. In *Advances in Neural Information Processing Systems*, 2022.
- Tianhe Yu, Garrett Thomas, Lantao Yu, Stefano Ermon, James Zou, Sergey Levine, Chelsea Finn, and Tengyu Ma. Mopo: Model-based offline policy optimization. In *Advances in Neural Information Processing Systems*, 2020.
- Tianhe Yu, Aviral Kumar, Rafael Rafailov, Aravind Rajeswaran, Sergey Levine, and Chelsea Finn. Combo: Conservative offline model-based policy optimization. In *Advances in Neural Information Processing Systems*, 2021.
- Zhuohao Yu, Zhiwei Steven Wu, and Adam Block. From curiosity to caution: Mitigating reward hacking for best-of-n with pessimism. *arXiv preprint arXiv:2604.04648*, 2026.

A PROOF SKETCH

For the diagnostic proposition, condition on the candidate set. If $\hat{i} = \arg \max_i (\mu_i + \epsilon_i)$, then candidates with larger realized residuals are more likely to be selected. In the equal-mean case the selected residual is exactly $\max_i \epsilon_i$, so its expectation is nonnegative and strictly positive for non-degenerate residuals. Unequal means can reduce but do not erase the mechanism when residual variance changes the argmax with positive probability. The proposition is used only as a mechanism sketch; the empirical artifacts measure selected residuals directly.

B IMPLEMENTATION DETAILS

The base posterior ensemble is generated from perturbed latent dynamics: gain, damping, hazard center, hazard strength, goal shift, value bias, and terminal bias vary across members. Candidate sequences are smooth controls with occasional high-excursion probes. Metrics include selected true return, posterior mean, posterior standard deviation, sampled score, regret to the in-set oracle, sample-over-mean residual, OOD mass, hazard mass, action energy, uncertainty percentile, and hazard-tail selection.

The learned bootstrap suite featurizes each action sequence using action moments, positive/negative action mass, cumulative-action proxies, velocity proxies, early/late sums, and time-weighted moments. Bootstrap ridge regressors are trained on true synthetic rollout returns from a narrower action distribution and evaluated under a wider distribution. This suite is a lightweight learned value-ensemble stress test, not a replacement for neural world-model benchmarks.

C CLAIM-EVIDENCE TABLE

Claim	Evidence artifact	Boundary
Posterior-sampled high- N selection can exploit epistemic tails.	full/summary; Figures 1–2.	Controlled latent dynamics, not hardware.
Failure persists across posterior stress conditions.	stress_summary; Figure 3.	Stress grid is synthetic and CPU-bounded.
Learned-model evidence exists beyond hand-coded posterior members.	learned_summary; Figure 4.	Bootstrap return ensemble, not a large neural world model.
High- N harm is paired and statistically reported.	paired_effects.	Normal-approximate CI over candidate problems.
Calibration is audited.	coverage.	Diagnostic coverage, not conformal safety.

D REVIEWER ATTACK SURFACE

Is this still synthetic? Yes. The paper says so, and the claim is not real-robot validation. The expanded evidence is not meant to hide the limitation; it tests whether the mechanism is robust to stress settings and a learned bootstrap ensemble.

Is pessimism the real contribution? No. The contribution is the posterior-tail diagnostic and evidence that the failure is measurable. Calibrated LCB is a sanity repair showing that the failure can be reduced with held-out residual calibration.

Does posterior mean already solve the base benchmark? Mostly yes, and the paper foregrounds that boundary. The risk is for deployments that use posterior-sampled, Thompson-like, optimistic, or otherwise upper-tail scores as the selection objective.

Does the learned suite prove generalization to large world models? No. It is a stronger controlled artifact, not a large-scale claim. It uses finite-data bootstrap uncertainty and evaluation shift to test whether the same selector pathology appears without hand-coded posterior members.

E COMPUTE BUDGET

The expanded suite uses 9 stress conditions, three seeds, bounded candidate pools, 24-member ensembles, and no model downloads. It is designed to run on CPU in roughly a minute on the development machine. This is why the evidence is broad across controlled conditions rather than heavy in a single large environment.

F SELECTOR DEFINITIONS

For a candidate set with posterior member predictions P_{ij} for candidate i and member j , the experiment evaluates the following deterministic or randomized scores:

$$\text{mean}(i) = \frac{1}{M} \sum_j P_{ij}, \quad (4)$$

$$\text{std}(i) = \sqrt{\frac{1}{M} \sum_j (P_{ij} - \text{mean}(i))^2}, \quad (5)$$

$$\text{sample}(i) = P_{i,J_i}, \quad J_i \sim \text{Uniform}\{1, \dots, M\}, \quad (6)$$

$$\text{Thompson}(i) = P_{i,J}, \quad J \sim \text{Uniform}\{1, \dots, M\}, \quad (7)$$

$$\text{UCB}(i) = \text{mean}(i) + \text{std}(i), \quad (8)$$

$$\text{LCB}(i) = \text{mean}(i) - \beta \text{std}(i). \quad (9)$$

The posterior-sampled selector uses an independently sampled member index per candidate. The Thompson-member selector uses one sampled member for the whole candidate set. This separation matters: coherent Thompson-style ranking tests a single posterior hypothesis, while per-candidate posterior sampling searches across posterior outliers.

The calibrated LCB uses a held-out calibration set. Let V_i be true synthetic return on calibration candidates. The scalar

$$\beta = Q_{0.85} \left(\frac{|\text{mean}(i) - V_i|}{\text{std}(i) + 10^{-8}} \right) \quad (10)$$

is clipped to a bounded interval and then reused on test candidate problems. Test true returns are not used by the selector.

G STRESS CONDITIONS

The posterior stress grid is designed to vary the mechanism rather than simply rerun the same benchmark. Each condition changes one or more knobs that affect posterior uncertainty or candidate extrapolation:

Condition	Spread	Blindspot	Action scale	Purpose
nominal	1.00	1.00	1.00	Base stress setting.
low blindspot	1.00	0.55	1.00	Reduced hallucinated tail value.
high blindspot	1.00	1.55	1.00	Stronger epistemic value bias.
low spread	0.65	1.00	1.00	Tighter posterior members.
high spread	1.45	1.00	1.00	Wider posterior dynamics.
gentle actions	1.00	1.00	0.75	Less extrapolative candidate generator.
wide actions	1.00	1.00	1.25	More extrapolative candidate generator.
hard shift	1.35	1.45	1.25	Combined posterior and generator shift.
calibrated shift	0.85	1.35	1.15	Moderate spread with stronger value bias.

The stress grid is not a substitute for real environments, but it closes a specific weakness of the controlled benchmark: the main conclusion no longer rests on one posterior parameterization. In each condition, the candidate oracle remains an in-set reference, so high regret for a deployed selector means the candidate pool contained good actions that the selector failed to choose.

H CALIBRATION COVERAGE DIAGNOSTICS

The calibration coverage file records, for each suite and condition, the rate at which the selected calibrated-LCB trajectory still has positive optimism gap. This is not a formal conformal guarantee; it is a falsification-oriented diagnostic. If calibrated LCB often selected trajectories whose posterior mean remained above true return, the repair would be unreliable even in the controlled setting. In the posterior stress suite, optimistic-miss rates remain small in the listed conditions, while the learned bootstrap suite is more fragile and exposes the cost of finite-data return modeling under evaluation shift.

This distinction is important for reviewer honesty. The artifact does not claim that calibration solves learned world-model shift. It shows that:

- posterior-sampled selection is highly fragile in the designed posterior stress conditions;
- calibrated LCB is consistently stronger there;
- under learned bootstrap shift, all non-oracle deployed selectors are weak, but LCB is still stronger than sampled-posterior selection on paired high- N problems.

I LEARNED BOOTSTRAP PROTOCOL

The learned suite uses synthetic data but removes hand-coded posterior members. For each seed, it samples training action sequences from a narrower action distribution and evaluates true returns under the latent dynamics. It featurizes each action sequence using action moments, cumulative-action

proxies, velocity-proxy summaries, high-excursion masses, and time-weighted moments. Then it fits a bootstrap ridge ensemble. At evaluation time, candidate actions come from a wider distribution. The same selector set is applied to the candidate-by-member prediction matrix.

This protocol is intentionally modest and reproducible. It does not require PyTorch, a GPU, a simulator license, or external data. The cost is that it is not a full neural world model. The benefit is that it creates a learned finite-data posterior-like ensemble whose uncertainty comes from bootstrap resampling rather than from hand-picked model-member parameters.

J HARSH REVIEWER CLOSURE

Reviewer attack	Closure
The paper is a short synthetic note.	The manuscript adds stress sweeps, learned bootstrap evidence, paired CIs, calibration coverage, and a detailed claim-evidence appendix.
The result is one cherry-picked posterior.	The stress grid varies posterior spread, blindspot strength, and action extrapolation in nine conditions.
The repair is just known pessimism.	The paper does not claim pessimism as an algorithmic invention; it uses LCB as a diagnostic repair proving the selected posterior tail is measurable and reducible.
Posterior mean already works.	The paper states this boundary and targets deployments that rank by posterior samples, optimistic scores, or upper-tail ensemble values.
No learned model is tested.	The bootstrap suite adds finite-data learned return ensembles under evaluation shift, with artifact-backed paired analysis.
No real robot evidence.	The title, abstract, limitations, and claim table avoid hardware claims. The contribution is a controlled diagnostic benchmark.
Statistics are missing.	Paired high- N effect sizes and 95% confidence intervals are generated in <code>paired.effects</code> .
Calibration might be hiding failures.	<code>coverage</code> records optimistic-miss rates and selected uncertainty under each condition.